SECTION A (40 Marks)

*(Attempt **all** questions from this **Section**.)*

Question 1

[15]

Choose the correct answers to the questions from the given options.

(Do not copy the questions, write the correct answers only.)

(i) A body is acted upon by two equal and opposite forces, that are **NOT** along the same straight line. The body will:

(a) remain stationary

(b) have only rotational motion

(c) have only rectilinear motion

(d) have both rectilinear and rotational motion

(ii) Which among the following is a **vector** quantity?

(a) work

(b) power

(c) energy

(d) moment of couple

(iii) What is the correct energy transformation during burning of a candle?

(a) heat → kinetic + potential

(b) heat → chemical + light

(c) chemical → heat + light

(d) mechanical → chemical + heat

(iv) When a ray of light passes from one optical medium to another, which of the following physical quantities does **NOT** change?

(a) Amplitude of the wave

(b) Frequency of the wave

(c) Wavelength of the wave

(d) Speed of the wave

(v) Which one of the following combinations is the correct **ascending order** of electromagnetic waves in terms of **wavelength**?

(a) gamma-rays, visible light, microwaves

(b) microwaves, visible light, gamma-rays

(c) gamma-rays, microwaves, visible light

(d) microwaves, gamma-rays, visible light

(vi) For a lever, a graph is plotted with load on Y-axis and effort on X-axis. Which of the following represents the **slope** of the graph?

(a) Mechanical advantage

(b) Velocity ratio

(c) 1 / Velocity ratio

(d) 1 / Mechanical advantage

(vii) For a real image formed by a convex lens, the ratio of **I: O = 2 : 5,** then the object is:
(I is the height of the image and O is the height of the object)

(a) between O and F

(b) beyond 2F

(c) at F

(d) between F and 2F

(viii) A ray of light is incident normally on a face of an equilateral prism. The ray gets totally reflected at the second refracting surface. **The total deviation** produced in the path of the ray is:

(a) 30°

(b) 60°

(c) 90°

(d) 120°

(ix) In a closed circuit containing a bulb and a cell, the electromotive force ($\boldsymbol{\varepsilon}$) and the terminal voltage ($\boldsymbol{V}$) is related as.
*(Given **I** is current and **r** is internal resistance.)*

(a) $V = \varepsilon + Ir$

(b) $V = \varepsilon - Ir$

(c) $V = \varepsilon \div Ir$

(d) $V = \varepsilon \text{ x } Ir$

(x) A metal piece of mass 5 g has thermal capacity 2.5 JK^{-1}. If the mass of the metal is tripled, then its **specific heat capacity** will be:

(a) 7.5 JK^{-1}

(b) 2.5 JK^{-1}

(c) 1.5 $Jg^{-1}K^{-1}$

(d) 0.5 $Jg^{-1}K^{-1}$

(xi) **Assertion (A):** As the level of water in a tall measuring cylinder kept under running tap rises, the pitch of sound gradually increases.

Reason (R): Frequency of sound is inversly proportional to the length of the water column.

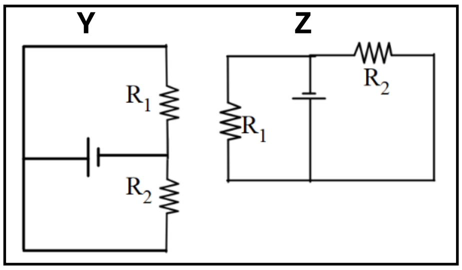
(a) Both (A) and (R) are true and (R) is correct explanation of (A).

(b) Both (A) and (R) are true and (R) is not the correct explanation of (A).

(c) (A) is true but (R) is false.

(d) (A) is false but (R) is true.

(xii) In the given circuits **Y** and **Z**, the resistors, $\mathbf{R_1}$ and $\mathbf{R_2}$, are connected in:



(a) series in both the circuits

(b) parallel in both the circuits

(c) parallel in **Y** and series in **Z**

(d) series in **Y** and parallel in **Z**

(xiii) A radioactive element **P** emits one α-particle and transforms to a new element **Q**. What will be the position of the element **Q** in the **periodic table**?

(a) One group to the left of **P**

(b) One group to the right of **P**

(c) Two groups to the right of **P**

(d) Two groups to the left of **P**

(xiv) Each of the substances given below is supplied with same amount of heat. Which one will attain the **highest** temperature if all have equal masses?

Substance	Lead	Aluminium	Copper	Iron
Specific heat capacity (cal/g °C)	0.031	0.21	0.095	0.115

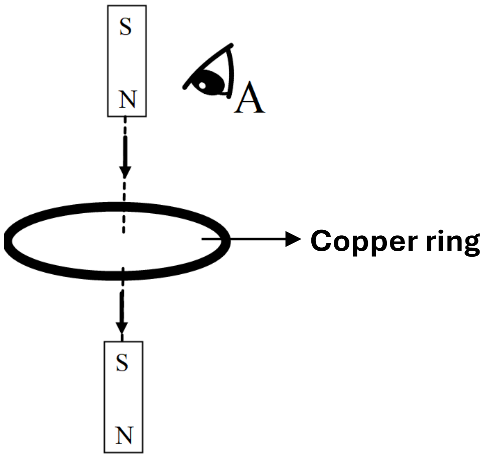
(a) Aluminium

(b) Copper

(c) Iron

(d) Lead

(xv) The following figure shows a small bar magnet falling freely through a copper ring. For the observer at **A**, the **direction of the induced current** will be:



(a) clockwise when magnet is above and below the ring

(b) anticlockwise when magnet is above and below the ring

(c) anticlockwise when magnet is above the ring and clockwise when the magnet is below the ring

(d) clockwise when magnet is above the ring and anticlockwise when the magnet is below the ring

MARKING SCHEME

(i)	(b) have only rotational motion
(ii)	(d) moment of couple
(iii)	(c) chemical → heat + light
(iv)	(b) Frequency of the wave
(v)	(a) gamma-rays, visible light, microwaves
(vi)	(a) Mechanical advantage
(vii)	(b) beyond 2F
(viii)	(b) 60°
(ix)	(b) $V = \varepsilon - Ir$
(x)	(d) 0.5 $Jg^{-1}K^{-1}$
(xi)	(c) (A) is true but (R) is false.
(xii)	(b) parallel in both the circuits
(xiii)	(d) Two groups to the left of **P**
(xiv)	(d) lead
(xv)	(c) anticlockwise when magnet is above the ring and clockwise when the magnet is below the ring

	Comments of Examiners	Suggestions for teachers
(i)	• A notable proportion of candidates answered the question incorrectly. • The concept of opposite forces was not clear to many candidates. • The most frequent error observed was that the balanced opposite forces would result in a stationary position of the object. • Some candidates suggested both linear and rotational movement, as pivot was not mentioned.	– Explain the concept of torque. – Highlight that two equal and opposite forces acting on a body will always cancel each other, resulting in no linear displacement but their torque on the body will be cancelled only if the two forces act along the same line. – Emphasise the distinction between net force and net torque. – Conduct a simple demonstration to show that non-pivoted object can also rotate. Example - rotating a pen to show how a net torque can cause rotation even without a fixed pivot. – Emphasise that the opposite forces are necessarily parallel to form a couple.

	Comments of Examiners	Suggestions for teachers
(ii)	• Many candidates answered it correctly. • Some candidates inaccurately chose one of the other three options due to lack of clarity regarding the fundamentals of vectors and scalars.	– Insist on learning the difference between a scalar and a vector quantity. – Provide relevant examples to explain that the product of two vectors is not always a vector.
(iii)	• Many candidates got this question correct. • Some candidates selected heat to chemical, and light. Perhaps, candidates assumed that the initial application of heat to light the candle is the primary energy input. • A few candidates also selected heat to kinetic and potential as linking it to energy changes during phase transitions and temperature increase.	– Encourage students to distinguish between energy states before and after a process. – Discuss different examples in the class that lead to the same energy conversions. – Use the demonstration in class to show energy transformations. Example- rubbing both hands together converts mechanical energy to heat energy. – Provide worksheet related to energy conversion for practice.
(iv)	• Most of the candidates answered it correctly. • Many candidates answered it incorrectly, either due to guesswork or lack of conceptual clarity.	– Explain the concept of refraction by demonstrating how a pencil placed in a glass of water appears to be bent. – Clearly mention the physical quantities that change and remain unchanged when light undergoes refraction.
(v)	• Many candidates answered it correctly as 'option a', however, some candidates marked the incorrect options also.	– Ensure that the students must know the electromagnetic spectrum in the increasing or decreasing order of the wavelength and frequency. – Use the mnemonics (Gamma) GIANT (X-rays) XYLOPHONES (UV) USUALLY (Visible) VISIBLE (IR) IN (Microwaves) MUSIC (Radio waves) ROOMS to memorise the correct order.
(vi)	• Most of the candidates answered it correctly as 'M.A.' Some candidates carelessly applied the formula of slope as $\frac{\Delta X}{\Delta Y}$ instead of the correct formula $\frac{\Delta Y}{\Delta X}$ and answered 'option d'.	– Provide a thorough explanation of the concept of finding the slope of a graph, along with its significance. – Lay stress on the name of the physical quantity that is derived from the slope of a graph. – Reinforce the importance of axis identification by providing examples.
(vii)	• Majority of the candidates found this question difficult.	– Discuss various cases of image formation in a convex lens with the help of suitable ray diagrams. – Elucidate the concept of magnification. – Highlight the importance of characteristics of the image in lens diagrams.

	Comments of Examiners	Suggestions for teachers
(viii)	• Many candidates could not perform well in this question as the basic concept of angle of deviation was not clear. • Some candidates misinterpreted the meaning of total deviation and answered it incorrectly. • The idea of total deviation, especially when there is a significant change in the light rays direction w.r.t the incident ray, needs more clarity.	– Help students visualise concept of total internal reflection and demonstrate it with the help of a glass of water and laser light in a dark room. – Emphasise that angle of deviation is angle between the original path of an incident ray and its final emergent path after passing through prism. – Instruct students to explicitly show and label the angle of deviation in the ray diagram wherever possible. – Focus on the idea of normal incidence and its implications on the trajectory of the ray.
(ix)	• Most of the candidates answered it correctly as option 'b'. However, some candidates got confused between + and – sign and wrote option 'a'. • Some candidates did guess work and wrote option 'c' or 'd'.	– Help students understand the relationship and derive the formula rather than memorising it. – Emphasise the distinction between electromotive force (ε) and terminal voltage (V), explaining the conditions under which they are equal or different (i.e. the role of internal resistance).
(x)	• Only a few candidates provided the correct answer. • The prevalent incorrect answer was 7.5, suggesting a misunderstanding where the given value was directly multiplied by 3, corresponding to the tripled mass. • Some candidates were not clear with difference between heat capacity and specific heat capacity. • Other incorrect options were randomly chosen.	– Explain the difference between heat capacity and specific heat capacity by taking suitable examples like water is used as a coolant in car radiator etc. – Reiterate that specific heat capacity being intrinsic property does not depend on the mass of the substance.
(xi)	• Many candidates wrote the wrong answers. • Some candidates lost marks as the question was not read carefully. • The candidates lack conceptual clarity related to the dependence of frequency of sound on air column.	– Focus on the difference between role of air column and water column in deciding the characteristics and propagation of sound. – Emphasise that frequency is inversely proportional to the length of the air column, not the length of the water column. – Instruct students to evaluate both the assertion and the reason individually before determining if the reason correctly explains the assertion or not.

	Comments of Examiners	Suggestions for teachers
(xii)	• A significant number of candidates did not answer this question correctly. • Some candidates got confused due to the change in the orientation of the diagram. Candidates seems to be not clear about circuit connections which deviates from the traditional representation.	– Teach students to analyse the circuit diagram carefully for series and parallel connections. – Clearly state the resistors that share the same potential difference at both ends are connected in parallel combination. – Provide practice of the circuit diagrams involving identification of connections of resistors.
(xiii)	• Many candidates attempted this question correctly. • Some candidates could not relate the change in atomic number with alpha emission to the change in position of the element in the periodic table. • A few candidates misinterpreted the left and right side of the periodic table.	– Emphasise the change in atomic and mass number of an element after alpha and beta emission and identity its position in the periodic table. – Provide explanation of the change in the number of protons and neutrons in the daughter nucleus after emission of the nuclear radiations through suitable explanation.
(xiv)	• Candidates who correctly answered the question focused on the conceptual analysis and identified the inverse relation between the specific heat capacity and rise in temperature.	– Emphasise inverse relationship that is for a given amount of heat energy supplied, the rise in temperature is inversely proportional to the specific heat capacity of the substance.
(xv)	• Many candidates failed to comprehend this question. • Some candidates applied Lenz's law correctly but failed to notice the side from which the current was observed. • Some candidates could not apply Lenz's law at all.	– Stress that the direction of the induced current (clockwise or anticlockwise) is always defined relative to the point of observation. The same current loop might appear clockwise from one side and anticlockwise from the other. – Use visual examples and demonstrations to illustrate different scenarios of changing magnetic flux and the resulting induced current direction. – Give diagram-based practice questions to apply Lenz's law for identifying the direction of the induced current. **Overall:** – Clarity in visualization of various ideas related to light and sound should be given to the students based on multiple demonstrations and by giving experience of analysing diverse ray diagrams. – The tendency toward rote memorization and mere learning of formulas by heart can be minimized through repeated discussions of case-based questions related to these

Comments of Examiners		Suggestions for teachers
		concepts. Students can be assigned such cases or real-life scenarios in groups and asked to deduce variable relationships on their own.

Question 2

(i) Complete the following by choosing the correct answers from the bracket: **[6]**

(a) In uniform circular motion the **centrifugal force** acts __________ *[towards the centre / away from the centre / along the tangential direction]*.

(b) Refractive index of a medium is **independent** of __________ *[temperature / angle of incidence / wavelength of light]*.

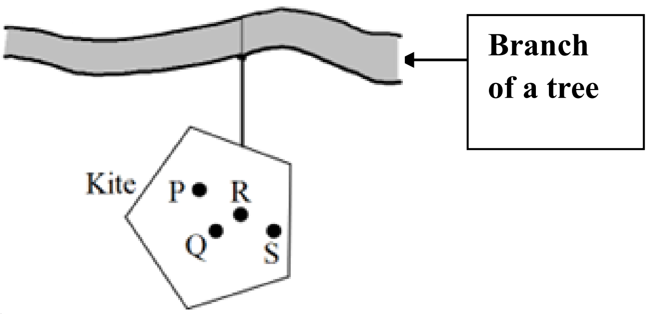
(c) Heat absorbed during **change of phase** depends on __________ *[mass / change in temperature / specific heat capacity]* of the substance.

(d) Emf of a cell is __________ *[greater than / less than / equal to]* the terminal voltage when the cell is in **open circuit**.

(e) In a step-up transformer the **turns ratio** is __________ *[more than 1 / less than 1 / equal to 1]*.

(f) The nuclear radiation with **lowest** ionizing power is __________ *[α / β / γ]*.

(ii) A **non-uniform** kite is hanging freely from the branch of a tree as shown. Study the figure and answer the following: **[2]**

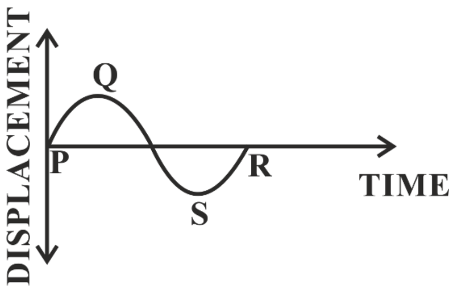


(a) **Fill in the blank.**

__________ (**P, Q, R** or **S**) is the most probable position of its centre of gravity.

(b) Support your answer to (a) with a reason.

(iii) The displacement-time graph of a sound wave produced by a vibrating wire is shown below. **[2]**



(a) How will you adjust the tension in the wire, to **reduce** the length of **PR**?

(b) Which characteristic of sound is affected by the reduction in the length of **PR**?

MARKING SCHEME

(i)	(a) away from the centre (b) angle of incidence (c) mass (d) equal to (e) more than 1 (f) γ radiation
(ii)	(a) R (b) When a body is in equilibrium the Centre of Gravity will lie vertically below the point of suspension / vertical line through R or plumb line through R passes through the point of support/ about the line of support /suspension the algebraic sum of the moments is zero
(iii)	(a) Increase the tension (b) Pitch / frequency

Comments of Examiners		Suggestions for teachers
2 (i)(a)	• Many candidates answered it correctly. • An equal number of candidates incorrectly answered, 'towards the centre', indicating confusion between 'centripetal' and 'centrifugal' force. • Few candidates incorrectly answered, 'along the tangential direction', possibly due to a misunderstanding of the velocity direction.	- Emphasise that centripetal force is a real force directed towards the centre of the circular path. - Clearly explain that centrifugal force is an apparent force experienced by an object moving in a circular path, directed outwards from the centre. - Demonstration method must be used to make students understand the concept with simple objects like a toy, or a tennis ball on a string.

Comments of Examiners		Suggestions for teachers
2 (i)(b)	• Most candidates answered it correctly. • Some candidates made errors in answering it. • Since $\mu = \frac{\sin i}{\sin r}$, few candidates thought that μ depends on the angle of incidence, hence answered it incorrectly.	- Stress that the angle of incidence is a variable that influences the angle of refraction according to Snell's Law, but it does not determine the refractive index of a material itself. - Highlight that μ is a characteristic property of a medium related to its optical density which depends on the temperature of the medium, colour / wavelength of light travelling through it.
2(i)(c)	• Many candidates were able to answer it correctly. • Several candidates misinterpreted 'heat absorbed' and related it to 'specific heat capacity' without considering 'change in state'. • Some candidates did guesswork in answering.	- Instruct students to read the question carefully specially the words given in bold. - Explain the concept of latent heat and highlight that the heat absorbed during a phase change does not cause a rise in temperature.
2(i)(d)	• Many candidates answered it correctly, several candidates could not answer it as well. • Some candidates were not clear with the concept of potential difference across the terminal of a cell in an open circuit is same as e.m.f. of a cell.	- Elucidate the concept of e.m.f., terminal voltage, and potential drop. - Explain the concept of internal resistance.
2 (i) (e)	• Most candidates wrote the correct answer. • Some vague answers were also observed.	- Elucidate the difference between the step-up and step-down transformer with the help of a model and explain the turn ratio.
2 (i) (f)	• Most candidates answered it correctly as γ radiations. • Some candidates wrote 'beta' and 'alpha' radiations that clearly indicates lack of clarity in understanding the properties of radioactive radiations.	- Teach the differences between the properties of nuclear radiations with the help of a comparison chart. - Relate the concepts to everyday experiences. For example: ➢ Alpha particles are like heavy bowling balls. ➢ Beta particles are like fast-moving marbles. ➢ Gamma rays are like powerful light beams.

Comments of Examiners		Suggestions for teachers
2 (ii)(a)	• Most incorrect option chosen was 'Q' instead of the correct answer 'R'. • Many candidates did not take into consideration that in a non-uniform mass distribution, position of C.G. is affected. • Very few candidates could relate to idea that 'the Centre of Gravity will lie vertically below the point of suspension' instead many of them might have assumed the geometrical centre as the COG.	- Explain the principle of finding C.G. of an irregular object. - Perform an activity using simple 3 d models or simuations can be helpful for better clarity on physical quantities and how they influence various physical system.
(ii)(b)	• Many candidates who selected R as the answer failed to give a correct reason, indicating a lack of conceptual clarity. • Some candidates wrote that the algebraic sum of the molecular moments of C.G. is zero as a reason but could not write the algebraic sum of the moments of the point of support is zero when C.G. lies vertically below the support. • Very few candidates got this part of the question correct.	- Emphasise that the algebraic sum of the moments of the molecules is always zero about C.G. but when the body is suspended freely then the sum of the molecular moments needs to be zero about the point of suspension and that happens when the C.G. is vertically below the point of suspension.
(iii) (a)	• Many candidates could not comprehend it correctly and failed to relate the length QR with the wavelength. • A considerable number of candidates commonly suggested physically tightening the wire or increasing the weight applied to it instead of tension of the wire. • Some candidates incorrectly stated that the wavelength should be reduced to achieve the desired outcome.	- Elaborate the fundamental characteristics of waves (e.g. frequency, wavelength, velocity, amplitude, tension for the waves on a string). - Explain the displacement time graph of a sound wave produced by a vibrating wire and indicate the wavelength, amplitude, and frequency on the graph. - Provide rigorous practice of identifying and labelling the graph.

Comments of Examiners		Suggestions for teachers
(iii)(b)	• Many candidates answered this part correctly. • Certain candidates were not clear that the characteristic of sound asked in the question was pitch, which is not the same as frequency. • Some candidates incorrectly wrote the answer as frequency or wavelength.	- Clearly explain that subjective properties of sound are the characteristics of a sound as perceived by a listener. - Connect subjective properties to their objective physical factors that influence the latter. - Demonstrate the characteristics of a sound with the help of a musical instrument to provide conceptual clarity.

Question 3

(i) A ray of light enters a rectangular glass slab submerged in water at an angle of incidence 55°. Does this ray undergo **total internal reflection** when it moves from water to glass? Justify your answer. *(The critical angle for glass-water interface is 54°.)* **[2]**

(ii) According to the **NEW** colour convention which colour of wire is connected to: **[2]**

(a) the metal body of the appliance

(b) the switch of the appliance?

(iii) (a) Which of the two, *alternating current* or *direct current*, produces a varying magnetic field when it flows through a conductor? **[2]**

(b) State the frequency of the alternating current supply in India.

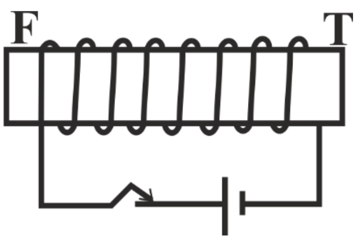
(iv) Calcuate the amount of heat absorbed by 200 g of paraffin wax to melt completely at its melting point. **[2]**

[Specific latent heat of fusion of paraffin wax = 146 Jg^{-1}]

(v) Copper wire is wound around a **steel** bar **FT**. Current is allowed to pass through the coil for some time and then the bar is removed. **[2]**

(a) Draw **only** the magnetised bar **FT** and mark its poles.

(b) Trace **two** magnetic lines of force around **FT** clearly indicating the direction.

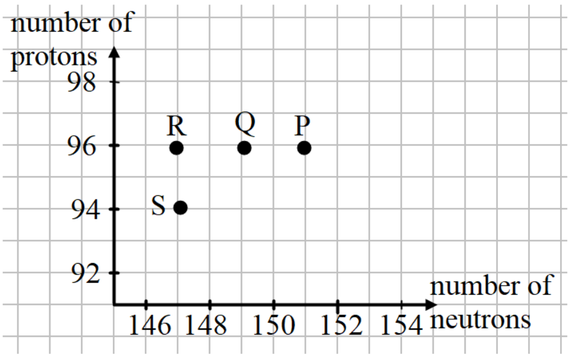


(vi) A current flows through a metallic conductor for a **long period** of time. State the change you would expect in its: **[2]**

(a) Resistance

(b) Resistivity

(vii) Curium is a radioactive element with the symbol $^{247}_{96}Cm$ named in honour of Madam Curie. The graph of **number of protons** vs **number of neutrons** for some elements are shown below: **[3]**

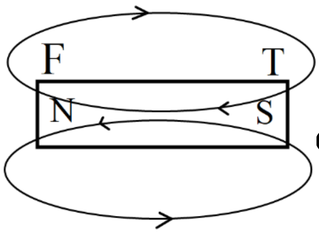
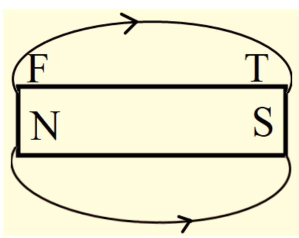
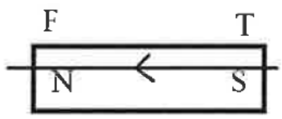


(a) Which point on the graph indicates the element *Cm*?

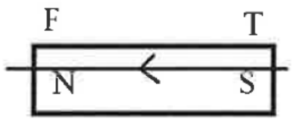
(b) Which point on the graph indicates daughter nucleus after *Cm* undergoes radioactive decay of 1 α followed by 2 β?

(c) State the mass number of the daughter nucleus.

MARKING SCHEME

(i)	No Total Internal Reflection don't take place when light travels from a rarer to a denser medium
(ii)	(a) Green or Yellow (b) Brown/ Red
(iii)	(a) AC/alternating current (b) 50 Hz
(iv)	H = ml = 200 x 146 = 29200 J
(v)	 Or  Or 
(vi)	(a) Resistance → Increases

	(b) Resistivity → Increases
(vii)	(a) P (b) R (c) 243

	Comments of Examiners	Suggestions for teachers
3(i)	• Most of the candidates answered it incorrectly. • Many candidates failed to recognise that the light ray was travelling from a rarer to a denser medium and concluded with the wrong answer. • Some candidates answered this question correctly.	- Teach the concept of TIR with examples along with the conditions for it to take place. - Include practice problems that involve light travelling both from rarer to denser and denser to rarer medium. - Give practice worksheets having variety of diagram-based questions involving the phenomenon of total internal reflection.
3 (ii)(a)	• A significant number of candidates correctly answered this direct question. • The primary source of error was confusion regarding the colour coding of electrical wires.	- Thoroughly explain the standard colour coding for electrical wires. - Use visual aids like actual wires or diagrams with clear labelling. - Emphasise the function and safety implications of each colour. - Reiterate to learn both old and new colour-coding of the wires and the functions of the wires in the household electric circuit.
3(ii) (b)	• A small fraction of candidates incorrectly identified wire connected to the metal body and the switch.	
3 (iii) (a)	• Most of the candidates answered it correctly. • A variety of incorrect options, like 'alternate' and 'altering' were used instead of the word 'alternating current' that was clearly mentioned in the question.	- Clearly state that varying current produces varying magnetic field around the conductor and constant current produces constant magnetic field around the conductor. 
3(iii) (b)	• Many candidates answered it correctly as '50 Hz' but many other candidates wrote the value '220 V'. • Some candidates wrote magnitude correctly but made a mistake in unit. • A confusion between the specifications of voltage and frequency has been repeatedly observed among candidates over the	- Encourage students to learn the value of frequency of alternating current and potential difference for household circuit. - Emphasise on the standard values along with its significance and implications on the practical systems.

	Comments of Examiners	Suggestions for teachers
	years. They tend to memorize '220' and '50' as values associated with domestic AC supply but fail to identify them distinctly as voltage and frequency.	
3(iv)	• Most of the candidates answered it correctly. • The common errors observed were: ➢ Missed or used the incorrect unit. ➢ Unnecessary mathematical conversions. ➢ Conversion of only one physical quantity.	- Make the students aware to substitute all the physical quantities in the same system of units in the formula. - The equations, relationships and characteristics should not be encouraged to be reproduced as such, instead multiple contextual examples and solving them should be the focus.
3(v) (a)	• Almost equal number of correct and incorrect answers were observed. • Some candidates changed the orientation of the bar magnet, but the labelling was correct. • Some candidates were unaware of the clock rule, so wrote the wrong polarity.	- Teach the clock rule with the help of demonstration in class. - Guide the students not to change the orientation of the figure given in the question.
3(v) (b)	The following variations were observed in the answers- • Correct polarities with correct direction of the magnetic field outside the magnet. • Incorrect direction of the magnetic field outside the magnet. • Direction of the magnetic field along the axis correctly and some of them made a mistake. • Direction of magnetic field above the bar magnet was marked correctly from north to south & below the magnet was marked incorrectly from south to north.	- Provide sufficient practice to students to draw diagrams with correct polarities and correct direction of magnetic field. - Stress on drawing the correct direction of magnetic field from north to south especially below the bar magnet.

Comments of Examiners		Suggestions for teachers
3(vi) (a)	• Many candidates answered it correctly. • Some candidates could not relate that longer time of the flow of current is related to the increase in the temperature and therefore results in increase in resistance.	- Explain the heating effect of current by giving examples from daily life. - Relate the effect of rise in temperature with increase in resistance by using kinetic theory of the molecules.
3(vi) (b)	• Many candidates did not pay attention to the word 'long period of time' . • Students have the tendency to associate resistivity as a constant and hence need not vary. They also might be familiar with the table showing constant value of resistivity for different materials. The emphasis of long period of time indicating Joule's heating effect and there by temperature variation of resistivity is missed out by many candidates.	- Clearly explain the difference between resistance and resistivity. - Emphasise the factors that affect resistance and resistivity. - Provide practice problems and worksheets to reinforce understanding.
3(vii) (a)	• Many candidates answered it correctly. • Some candidates could not analyse and interpret the graph.	- Encourage students to practice interpreting graphs presented in specimen papers and competency -focused questions (CFQs). - Explain how to calculate of number of neutrons from atomic number and mass number
3(vii) (b)	• This part was correctly answered by the candidates who got part (a) of this question right. • Some candidates failed to read the questions thoroughly. Instead of number of 'protons' and 'neutrons', they considered them as 'atomic number' and 'mass number'. • Some candidates who answered it as 'S' considered the change in atomic number due to alpha emission.	- Encourage students to formulate an equation to solve the problem. - Provide practice questions, involving alpha and beta emission including reverse calculation of mass number (A) and atomic number (Z) in form of graphical data. - Clearly state that one alpha emission and two beta emissions produce an isotope of the parent nucleus with mass number less by 4. - Make it clear to the students that in this case the total reduction in the number of neutrons in the daughter nucleus is 4.

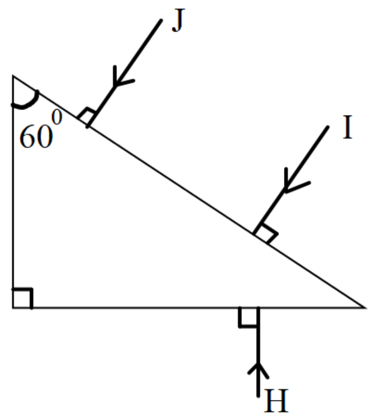
Comments of Examiners		Suggestions for teachers
3(vii) (c)	The following errors were observed: ➢ Misidentified axis for the number of neutrons by mass number. ➢ Represented the mass number of the daughter nucleus as alpha and beta decay reactions. ➢ Wrote mass number with the units.	- Instruct students to thoroughly examine the graph before drawing conclusions. - Emphasise that the axes are key to understand the graph's meaning. - Ensure sufficient practice of completing the nuclear equations.

SECTION B (40 Marks)

*(Attempt **any four** questions from this **Section**.)*

Question 4

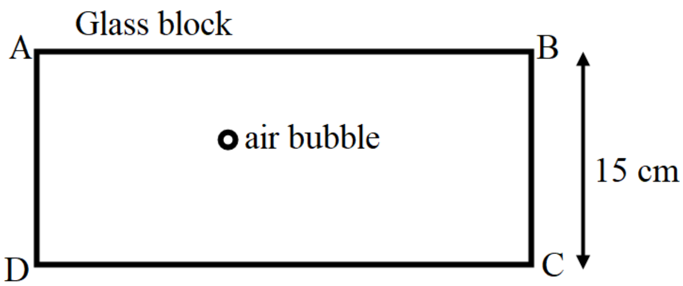
(i) **[3]**



(a) Out of the three rays (**I, J, H**) shown in the diagram, which ray will suffer **Total Internal Reflection** while inside the prism? *(Critical angle of the prism is 42°.)*

(b) Copy the diagram to complete the path of the ray which you have named in (a) till it comes out of the prism.

(ii) A rectangular glass block of refractive index 1.5 has an air bubble trapped inside it as shown in the diagram. When seen from the surface **AB,** it **appears** to be 4 cm deep. **[3]**



(a) Calculate the **actual depth** of the air bubble from the surface **AB**.

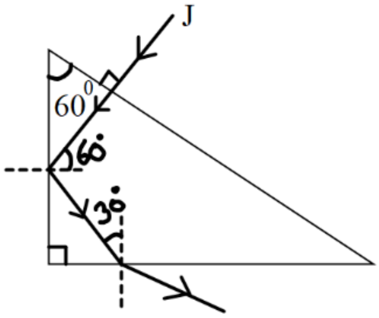
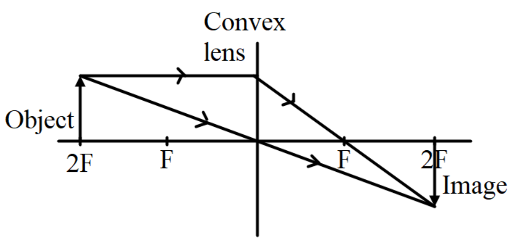
(b) For which colour of light, blue or yellow, the apparent depth will be **greater**?

(c) Turning the glass block upside down, **DOES NOT** change the apparent depth of the air bubble. State **True** or **False.**

(iii) (a) An object is placed at **2F** position of a convex lens. Draw a ray diagram showing the formation of the image. **[4]**

(b) How will the size of the image change if we, **ONLY** replace the lens in the above arrangement with another lens of a **greater focal length**?

MARKING SCHEME

(i)	(a) J (b) 
(ii)	(a) $\mu = \frac{RD}{AD} \therefore 1.5 = \frac{RD}{4} \therefore RD = 6\ cm$ (b) Yellow/ Y (c) False / F
(iii)	(a)  (b) Magnified / same size

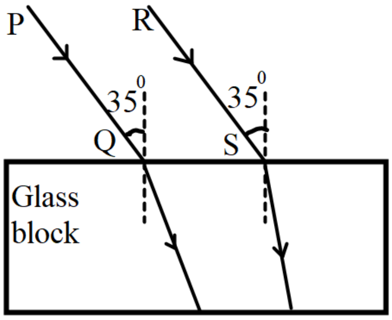
Comments of Examiners		Suggestions for teachers
4 (i) (a)	• Majority of the candidates accurately identified the ray 'J'. • Some candidates got confused between ray 'J' and 'I'. • Some candidates drew the correct ray diagram instead of naming the ray.	- Teach and explain the method on how to complete the path of the rays by calculating the angle of incidence at the inner surface using basic geometry. - Give practice of completion of path of the ray by taking triangular glass prism with interior angles as - 42°,48°, 90°/44°,46°,90° etc. - Encourage students to draw the diagrams neatly. - Prisms of different angles and light rays entering and leaving the surface with clarity in different angles (incident, reflection, refraction etc.) can be given to the students.
4 (i)(b)	• Some candidates drew the diagram correctly; however, some made it partially correct. • Some common errors that were observed - ➢ Forgot to draw arrows on the ray. ➢ Not mentioned angle of incidence at the new surface. ➢ Mentioned angle of incidence with the surface of the prism instead of normal.	- Insist upon marking the arrows in the ray diagram. - Provide rigorous practice in drawing a wide variety of ray diagrams related to TIR and critical angle to enhance confidence and understanding among students. - Help students to complete the path of the ray of light by calculating the angle of incidence at every new surface using basic geometry when the ray strikes the surface.
4(ii)(a)	• Most of the candidates could do this question correctly. • Some candidates lost marks due to incorrect unit.	- Explain the difference between real depth and apparent depth along with the shift in the position of the image. - Demonstrate real and apparent depth with the help of a glass slab or a coin placed in a glass of water. - Include questions where observer is in a denser medium and observing an object in a rarer medium.
4 (ii)(b)	Some candidates made mistakes as they did not keep in mind that the question was in terms of apparent depth and bending while some other candidates relied on guesswork. Majority of the students can learn the formulae and substitute it. But when it comes to the extension of the familiar scenario, as in this case where the candidates could not connect it to	- Encourage students to learn the relation between refractive index and the wavelength of light. - Ensure that learning formulas and equations should not be a mechanical process, students should be able to interpret those relations, for instance the idea of real depth and apparent depth in different contexts should be demonstrated through diagrams, simulations, videos etc.

	Comments of Examiners	Suggestions for teachers
	change in real and apparent depth, the gap in conceptual clarity becomes evident.	and students should solve various problems situated in these variety of scenarios.
4(ii)(c)	• Many candidates answered it correctly. • Some candidates could not comprehend the meaning of 'turning the glass upside down' and made mistakes.	- Reinforce the concept of real and apparent depth by taking real life examples. - Create and provide practice of situation-based questions to the students. - Clearly state that the ratio between real depth and apparent depth is constant therefore, when real depth changes, apparent depth also changes.
4 (iii)(a)	Majority of candidates missed full marks due to : • imperfections in the drawing. • indication of direction of light with arrows. • not maintaining a uniform scale on both sides of the lens. • object not drawn at 2F. • dotted lines used for a ray of light.	- Reinforce the rules for drawing ray diagrams by repeating them during practice of ray diagrams. - Instruct students not to use arrow for virtual rays. - Demonstrate different cases of an object placed at various positions in front of the convex lens in lab. - Insist on drawing continuous lines for real rays/image and dotted lines for virtual rays/image. - Provide practical opportunities to manipulate various optical systems using lens and its combinations.
4 (iii)(b)	• Several candidates were unable to answer this part correctly due to lack of analysis of the question and missed the hints given in bold. • Some candidates incorrectly changed the position of the object.	- Include the explanation of factors that affect the size of the image formed by the lens. - Perform an activity in the classroom where students observe the size of the image formed when focal length of the lens is changed.

Question 5

(i) An object is placed in front of a concave lens at a distance of 45 cm from it. If its image is formed at a distance of 30 cm from the lens, calculate the focal length of the lens. **[3]**

(ii) Two rays **PQ** and **RS** are incident on a rectangular glass block as shown in the diagram. Observe the diagram and answer the questions that follow. **[3]**



Which of these two rays will:

(a) have **greater** lateral displacement on emerging out of the block?

(b) travel with **greater** speed in the block?

(c) scatter **more** in the atmosphere?

(iii) (a) Name the **radiations**: **[4]**

1. for which a quartz prism is used to study the spectrum.
2. which are used in remote sensing devices.
3. which are used in traffic signals in India.

(b) Name **one** property **common** to all electromagnetic radiations.

MARKING SCHEME

(i)	u = –45 cm, v = –30 cm $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \therefore \frac{1}{f} = \frac{1}{-30} + \frac{1}{45}$ $\therefore \frac{1}{f} = \frac{-3+2}{90} = -\frac{1}{90} \therefore f = -90\ cm$
(ii)	(a) RS/ only R /only S / second ray (b) PQ / only P /only Q/ first ray (c) RS/ only R /only S / second ray
(iii)	(a) 1. Ultra-violet radiations/UV rays

	2. Infra-Red radiations/ IR radiations / micro Waves / radio Waves 3. Visible radiations/light /Any colour (R, G, Y, O, amber) (b) No medium required, or all can refract / reflect or travel with same speed in vacuum / air

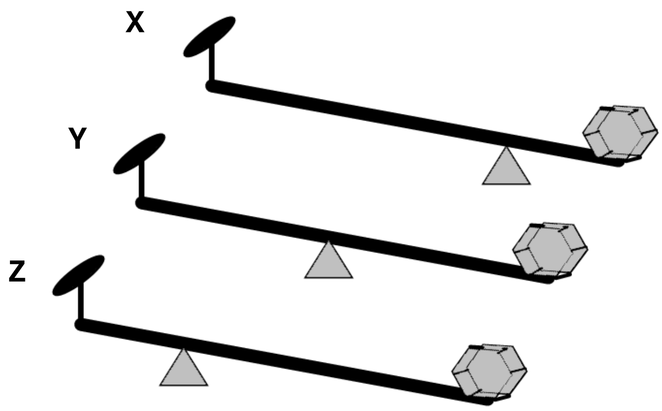
	Comments of Examiners	Suggestions for teachers
5 (i)	Many candidates got this numerical correct. However, following errors were observed: • Inability to identify the object and image distance properly. • Applied the wrong sign convention/ substitution. • Used the incorrect formula.	- Explain the rules and sign convention for numericals based on lens formula. - Guide students to apply sign convention at the time of substitution to avoid calculation errors. - Insist on learning the correct position and the characteristics of the image formation. - Provide practice questions related to lens formula.
5 (ii)(a)	• Many candidates could not relate the angle of refraction to the lateral displacement of the ray of light. • Some candidates answered it correctly. • Some candidates failed to notice the extent of refraction of the refracted rays for the same angle of incidence.	- Encourage students to minutely observe the diagram and draw conclusions from it. - Insist on breaking down the given question into smaller, manageable parts, each relating to one of the identified core concepts. - Encourage students to answer these sub-questions individually before trying to connect them.
(ii)(b)	• Many candidates failed to understand that more bending corresponds to more decrease in the speed of light in that medium. • Only some of the candidates managed to give the correct answer.	
(ii)(c)	Some candidates answered it correctly, but some could not identify that the ray which bends more has a shorter wavelength and therefore scatters more.	
(iii)(a)	• Most candidates gave the correct answers. • Some candidates were confused between 'ultra-violet' and 'ultrasonic' in part 1. • Some confused 'infrared' with 'infrasonic waves' in part 2.	- Encourage students to learn the applications and uses of infrared and ultraviolet radiations. - Conduct regular quizzes to revise the uses. - Ensure to include the type of prism used for obtaining the spectrum of infrared and ultraviolet radiations along with proper reasoning.

	Comments of Examiners	Suggestions for teachers
(iii)(b)	• Most of the candidates answered this question correctly. • A few candidates got confused between the properties of 'electromagnetic' and 'radioactive radiations.	- Lay stress on learning the common properties of electromagnetic radiations.

Question 6

(i) Akash takes a **uniform** meter scale and suspends a weight of 2 N at one end **'X'** and a weight of 5 N on the other end **'Y'**. He then balances the ruler horizontally on a knife edge placed at 70 cm from **X**. Draw a diagram of the arrangement and calculate the weight of the ruler. **[3]**

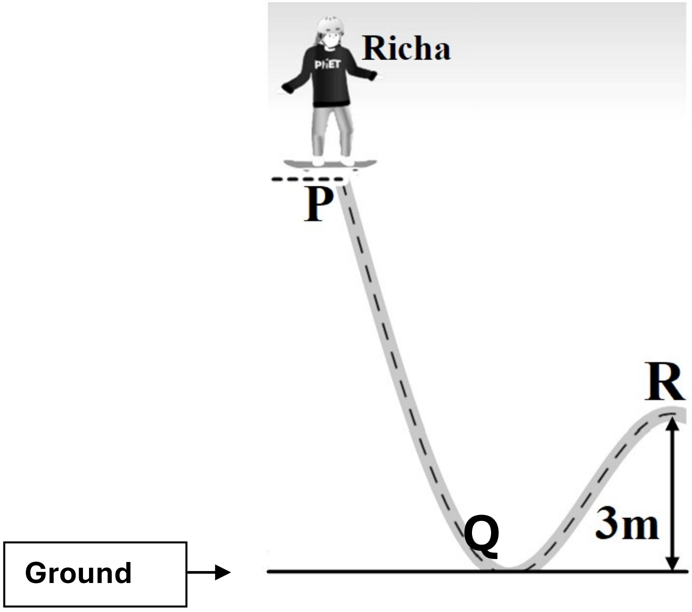
(ii) Three levers **X, Y, Z** of **equal lengths** are shown in the diagram. **[3]**



(a) Which class of lever do these belong to?

(b) Among these (**X, Y** or **Z**) which one will give the **maximum** mechanical advantage? Justify your answer.

(iii) **[4]**



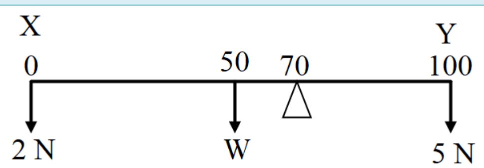
Richa weighing 40 kgf leaves point **P** on her skateboard and reaches point **Q** on the ground with velocity 10 ms^{-1}. Calculate:

(a) the kinetic energy of Richa at point **Q**.

(b) the vertical height of point **P** above the ground. *(Use **g** as 10 m/s^2 and neglect friction)*

(c) the kinetic energy of Richa at point **R.** ***(****While moving from **Q** to **R**, she loses 500 J of energy against friction.)*

MARKING SCHEME

(i)	 By pr. of moments (2 x 70) + (W x 20) = 5 x 30 ∴ 20 W = 150 – 140 20 W = 10 ∴ W = 10/20 = 0.5 N
(ii)	(a) First Class lever / I (b) Lever X EA > LA

(iii)	(a) $K = \frac{1}{2} mv^2 = \frac{1}{2} \times 40 \times 100 = 2000$ J (b) $\frac{1}{2} mv^2 = mgh \therefore 100/2 = 10 \times h \therefore h = 5$ m Or $mgh = 2000 \quad \therefore 40 \times 10 \times h = 2000 \quad \therefore h = \frac{2000}{400} = 5m$ Or (using formula $v^2 = 2gh$] (c) $U = mgh = 40 \times 10 \times 3 = 1200$ J K at R = 2000 – 500 – (1200 or 40 × 10 × 3) = 300 J

Comments of Examiners		Suggestions for teachers
6 (i)	Candidates show a lack of conceptual understanding on principle of moments. A variety of errors were seen in the answers- • Failed to comprehend the question. • Could not draw forces at the end of the scale. • Incorrectly marked the position of the fulcrum. • Lacked clarity about the fulcrum being closer to a larger weight and away from the smaller weight for the ruler to be in equilibrium. • Neglected to include the moment of force due to the weight of the ruler while formulating the equations for clockwise and anticlockwise moments.	- Emphasise drawing the diagram with the numerical for better understanding. - Clearly highlight taking perpendicular distance from the fulcrum for the calculation of moment of force. - Explain the position of fulcrum and identification of C.G. point. - Clearly state the position of the centre of gravity to be taken in the centre if the metre rule is uniform. - Provide hands-on experience to students to try balancing scales using different masses and fulcrum positioning. They can try to do a comparative study of their observation and the theoretical calculations using the equation of moments. Further discuss CG, principle of moments, distribution of masses etc.
(ii)(a)	• Majority of the candidates answered it correctly. • A significant number of candidates failed to depict that all three diagrams belonged to same class of lever.	- Use real-life examples to explain the kinds of levers with respect to the position of the fulcrum, load, and effort.
(ii)(b)	• Some candidates answered it as highest effort arm, but the comparison was needed between effort arm and load arm of the same lever.	- Clearly explain the term mechanical advantage of a lever using relevant examples. - Give sufficient questions related to M.A. in terms of load arm and effort arm.

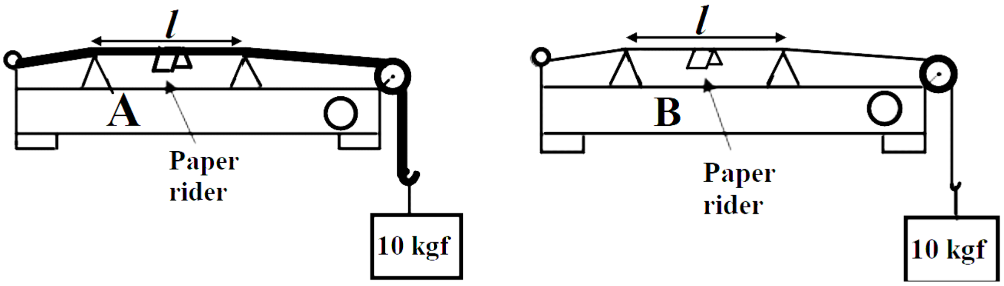
	Comments of Examiners	Suggestions for teachers
(iii)(a)	• Many candidates calculated K.E. correctly. • Errors were observed due to: – Incorrect conversion of kilogram-force (kgf) to newton (N). – Erroneous conversion of kilogram-force (kgf) to gram (g). – Calculation mistakes.	- Lay stress on the importance of converting the non-SI units to SI units. - Instruct students to note the numeric values carefully from the question.
(iii)(b)	• Many candidates could calculate the height by using the principle of conservation of energy. • Some candidates made a calculation error in the previous part, so the answer for this part was incorrect, though the concepts were clear. • Some of them made errors in substituting the incorrect values that were due to the wrong interconversion of the units.	- Discuss common errors in the class and encourage them to analyse their own errors and correct them. - Provide hand holding in the conceptualization of the given scenario and connecting it to the principle of conservation of energy. For this, multiple such scenarios from real life and other examples should be discussed.
(iii)(c)	• Many candidates were not able to reach the final answer due to a lack of clarity of the concept of 'energy conservation'. • Some candidates calculated U at point R and subtracted it from total K at Q, but failed to subtract the loss of energy to arrive at the correct answer. • Many candidates equated U and K at R and made a mistake in calculation.	- Provide more practice of diagram-based problems in the class so that students can apply the concept of the law of conservation of energy without any ambiguity. - Reiterate that the total mechanical energy remains constant only in the absence of friction.

Question 7

(i) Draw a block and tackle system of pulleys with **velocity ratio equal to 3**. **[3]**

(ii) A submarine in the sea, sends ultrasonic ping and a stopwatch is started simultaneously. The stopwatch stops on receiving the reflected wave from an obstacle and reads **1 minute 40 seconds**. Calculate the distance of the obstacle from the submarine. *(Speed of sound in water 1500 ms^{-1}.)* **[3]**

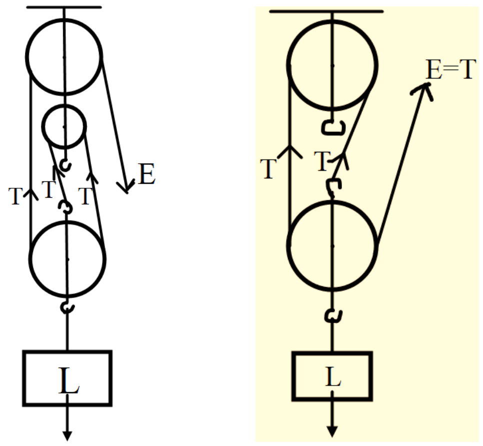
(iii) The diagrams given below show two sound boxes **A** and **B** with wires of the same material and **same** length ***(l)*** and tension *(10 kgf)* but **different** cross-sectional areas. Simultaneously, vibrating tuning forks of frequency 300 Hz are placed on the boxes **A** and **B**. The paper rider falls off in case of **B** but not in case of **A**. **[4]**



(a) **Name** and **explain** the phenomenon responsible for the falling off of the paper rider in **B**.

(b) The wire **A** resonates with a tuning fork of frequency ***'f'***. Is ***'f'*** *greater than, less than or equal to* 300 Hz? Justify your answer.

MARKING SCHEME

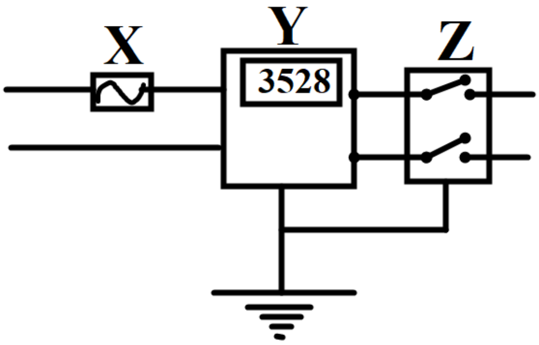
(i)	
(ii)	$V = \frac{2d}{t} \therefore 1500 = \frac{2d}{100}.$ $\therefore d = 1500 \times 50 = 75000\, m(= 75\ km)$
(iii)	(a) Resonance Frequency of tuning fork matches with natural frequency of wire on B /increases its amplitude. (b) f is less than 300 Hz. This is because mass per unit length of wire A is greater than mass per unit length of B and $f \propto \frac{1}{\sqrt{mass\ per\ unit\ length}}$.

<table>
<tr><th colspan="2">Comments of Examiners</th><th>Suggestions for teachers</th></tr>
<tr><td>7 (i)</td><td>• Very few candidates could draw the diagram correctly.
• The following errors were observed:
➢ Fixed and movable blocks were not clear.
➢ Pulleys in the fixed block were not connected.
➢ Pulleys and hooks were connected from the rim.
➢ Loose strands were drawn.
➢ Main support was not drawn.
➢ Either L, E, or T was not marked or, in some cases, was wrongly marked.
➢ Marked tension as T_1, T_2 &T_3 without understanding that the tension in all strands of a single tackle was the same.</td><td>- Explain the relation between V.R. value and the number of pulleys in a block and tackle system.
- Highlight that if V.R. is even, the number of pulleys in fixed and movable pulleys are same. And if V.R. is odd, the number of pulleys in the upper fixed block is one more than the lower movable block.
- Guide the students to draw arrows to indicate tension in each strand.
- Provide sufficient practice to draw the block and tackle system with different values of V.R.
- The concept of velocity ratio should not be limited to its formula. While teaching this concept, multiple scenarios using diagrams and simple models should be used to transact the idea of the distance travelled by effort and load.</td></tr>
<tr><td>7 (ii)</td><td>• Many candidates did the numerical correctly.
• However, some errors were observed-
➢ Conversion of minutes to seconds.
➢ Not considered reflection of sound during calculation.
➢ Common calculation errors.
➢ Incorrect unit.</td><td>- Emphasise the importance of unit conversion for obtaining the correct answer.
- Guide students to read the question carefully in terms of reflection of sound and use the formula accordingly.
- Encourage students to recheck their answer while performing mathematical calculations.</td></tr>
<tr><td>7 (iii)(a)</td><td>• Most of the candidates answered it correctly as ‘resonance’ but failed to explain the phenomenon correctly.
• Many candidates missed the crucial detail about the equality of specific frequencies.
• Some of the common errors included were:
➢ Stating that the frequency of the tuning fork is equal to the frequency of the paper rider.
➢ Mentioning the frequency of the tuning fork is equal to the frequency of the wooden box.</td><td>- Clearly differentiate between the driving frequency (of the tuning fork or external force) and the natural frequency.
- Explain the term resonance by giving examples from daily life.
- Explicitly address the common errors of equating the tuning fork’s frequency to the paper rider or wooden box.</td></tr>
</table>

Comments of Examiners		Suggestions for teachers
7 (iii)(b)	• Most of the candidates answered it incorrectly. • Many candidates failed to notice the different thickness of the strings in the diagram.	- Provide practice of diagram-based questions. - Explain the factors affecting the frequency of the string with the help of a demonstration. - Demonstrate how the frequency of a rubber band changes when it is stretched to different extent.

Question 8

(i) The diagram shows wiring in a meter room of a building. **[3]**

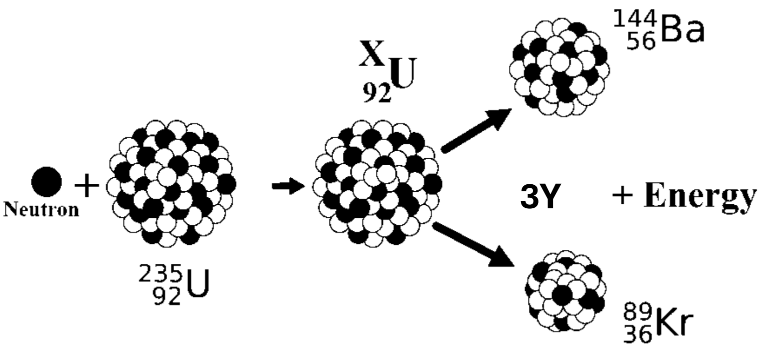


(a) What is the current rating of device **X**?

(b) What is the difference between the switch **Z** shown in the diagram and the switches you use to operate different appliances at home?

(c) What is the unit of the physical quantity displayed in **Y**?

(ii) Study the diagram given below and answer the questions that follow: **[3]**

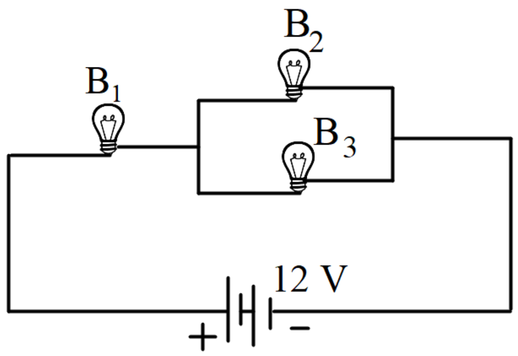


(a) Name the process depicted in the diagram.

(b) What is the value of **X**?

(c) Identify **Y**, the missing product of the reaction.

(iii) **[4]**



Three identical bulbs **B**$_1$, **B**$_2$, and **B**$_3$ each of power rating 18 W, 12 V are connected to a battery of 12 V.

(a) Calculate:

1. the resistance of each bulb
2. the current drawn from the cell

(b) If the bulb **B**$_3$ is removed from the circuit, then will the brightness of the bulb **B**$_1$ *increase, decrease, or remain the same*?

MARKING SCHEME

(i)	(a) 50 A (b) Switch Z is present in live as well as neutral wire / double pole switch whereas the switch in the house is present only in live/ single pole switch. (c) kWh/ kilowatt hour
(ii)	(a) (Nuclear) fission (b) 236 (c) (3) neutrons / n / ${}_0^1n$
(iii)	(a) 1. $R = \frac{V^2}{P} = \frac{12 \times 12}{18} = 8\,\Omega$ 2. $R_P = \frac{8 \times 8}{8 + 8} = 4\,\Omega \therefore R = 4 + 8 = 12\Omega$ $V = IR \therefore 12 = I \times 12 \therefore I = \frac{12}{12} = 1\,A$ (b) Decrease / less

Comments of Examiners		Suggestions for teachers
8(i) (a)	• A few candidates answered it correctly. • A variety of random answers were observed. • Some candidates wrote rating of the main fuse of the ring distribution system instead of the rating of the company's fuse.	- The students should be asked to observe the meter room and difference in the ratings at various stages of supply should be discussed. - Explain the connections by cable from electric pole to distribution box with the help of a model or a video. - Clearly mention that the rating of the fuse depends on the load. - Highlight that the main switch is a double pole switch. - Provide practice of variety of diagram-based questions and mention the common mistakes made by students in them. Involve problems that combine multiple concepts.
8(i)(b)	• A significant number of candidates correctly identified the switch. • Many candidates only described switch Z and missed out the explanation for the single-pole switch. • Some candidates incorrectly considered Z as a 'dual control switch', indicating a possible misunderstanding of the specific terminology. • Some of the candidates even took it as a 'MCB'.	- Explain the difference between a double-pole switch and a dual control switch and their functions. - Use clear diagrams illustrating the internal mechanisms and wiring of both a single-pole switch and a double-pole switch. - Encourage students to find the key differences between a double-pole switch and a single-pole switch with the help of a physical model.
8(i)(c)	• Majority of candidates provided the correct unit, which is kilowatt hour (kWh). • Some candidates could not interpret the diagram correctly. • Some candidates even mentioned the incorrect units.	- Clearly distinguish between power (measured in kW) and energy (measured in kWh) along with their units. - Integrate the reading of actual electric meters into classroom activities. - Provide varied examples and practice problems where students must determine both the power consumption and the energy used for various appliances over different time intervals.

	Comments of Examiners	Suggestions for teachers
8(ii) (a)	• Many candidates answered it correctly. • Some candidates named the process incorrectly as 'nuclear fusion' and 'radioactive reaction' while some wrote 'chain reaction' that was an incomplete answer.	- Explain the difference between nuclear fission and fusion reactions with the help of a comparative chart.
(ii)(b)	• Majority of candidates correctly identified the mass number of the intermediate 'uranium' isotope. • However, some candidates incorrectly wrote the value of X as '235', since they missed adding the mass of a neutron. • Due to the lack of knowledge of balancing the nuclear equation, the value of mass number was written wrongly.	- Provide sufficient examples and practice of finding the atomic and mass number of an element with the help of nuclear reactions. - Highlight that the law of conservation of mass is obeyed in a nuclear reaction. - Review the process of neutron bombardment step-by-step. - Emphasise that when a nucleus captures a neutron, its mass number increases by one, but its atomic number remains the same.
(ii) (c)	• Majority of candidates correctly determined the unknown particle as a neutron. • However, some candidates incorrectly identified the missing particle as either an 'alpha' (α) or a 'beta' (β) particle or even 'gamma' (γ), suggesting a confusion with radioactive decay emissions. • Some of the candidates identified a neutron but used the wrong notation to represent it.	- Utilise clear and distinct diagrams illustrating both nuclear fission and radioactive decay processes. - Emphasise the input (e.g., neutron for fission) and the output particles for nuclear fission and nuclear fusion reactions.
(iii)(a1)	• Majority of the candidates answered this subpart correctly. However, some candidates failed to use the relation $R= V^2/P$ and used the other alternatives. • Some candidates could not comprehend it properly and made incorrect calculations.	- Ensure that students understand the interconnectedness of power, voltage, current, and resistance through the fundamental formulas (P=IV, V=IR). - Give practice in finding the resistance of resistors in series and parallel.

Comments of Examiners		Suggestions for teachers
(iii)(a2)	• Many candidates were confused in understanding the series and parallel combination of resistors. • Several candidates wrote an incorrect expression. Hence, in the last step, the mathematical result was incorrect. • Some candidates got confused with the current rating and the current in the circuit, while some other candidates directly used the value in the formula.	- Clearly state that when the supplied voltage is different from the rated voltage, then current and power change, but the resistance of the appliance remains the same. - Encourage the students to understand and learn the formulas.
(iii)(b)	• Many candidates could not answer it correctly. They were not aware that removing one of the resistances from the parallel combination resulted in net increase in resistance. • Very few candidates could answer it correctly that on removing a bulb in this circuit configuration would lead to decrease in the brightness of the remaining bulbs.	- Explain Ohm's Law and verify it experimentally. - Demonstrate relation between the resistance of the bulb and its brightness in laboratory. - Students should be given opportunity to analyze multiple real circuits practically, through virtual labs and through diagrams. - They should be asked to design circuits to meet various voltage and current requirements through combination of resistors.

Question 9

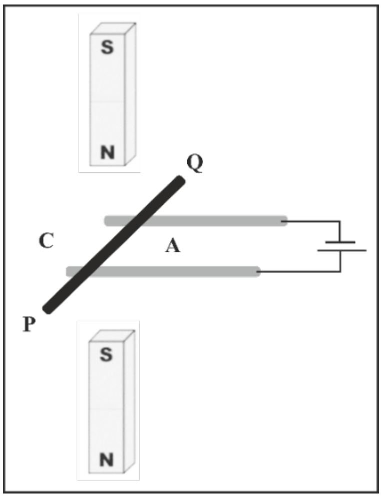
(i) 30 g of ice at 0°C is used to bring down the temperature of a certain mass of water at 70°C to 20°C. Find the mass of water.*[Specific heat capacity of water = 4.2 $Jg^{-1}°C^{-1}$ and specific latent heat of ice = 336 Jg^{-1}.]* **[3]**

(ii) (a) A certain amount of heat will warm 1 g of material **X** by 10°C and 1 g of material **Y** by 40°C. Which material has **higher** specific heat capacity? **[3]**

(b) Which material, **X** or **Y**, would you select to make a calorimeter?

(c) The specific heat capacity of a substance remains the **same** when it changes its state from solid to liquid. State **True** or **False.**

(iii) A copper rod **PQ** carrying current is kept in a magnetic field as shown in the diagram. **[4]**



(a) The copper rod **PQ** will move towards **C**. State **True** or **False**.

(b) **Name** the law used to determine the direction of motion of **PQ**.

(c) What will be the effect on the force experienced, if the rod **PQ** is replaced by another copper rod of **same** length but of **greater** cross-sectional area?

(d) Justify your answer in (c).

MARKING SCHEME

(i)	By pr. of mixtures $ml + mc\,\Delta t = mc\,\Delta t$ $30(336 + 4.2 \times 20) = m \times 4.2 \times 50$ $\therefore 30 \times 4.2(80 + 20) = 4.2 \times 50 \times m$ $\therefore 30 \times 100 = 50 \times m$ $\therefore m = 60$ g
(ii)	(a) X (b) Y (c) False /F
(iii)	(a) True / T (b) (Fleming's) Left-hand rule/ right hand palm rule (c) Force increases (d) As resistance decreases / current increases

	Comments of Examiners	Suggestions for teachers
(i)	• Many candidates calculated it correctly. • Some common errors were observed - ➢ Considered wrong equation that is mcΔt = ml or mcΔt = mcΔt. ➢ Converted the values of c and l in SI unit while leaving mass in CGS unit. ➢ Errors in calculations.	- Clarify that even if c and l are substituted in SI and mass in CGS unit the answer will be correct because conversion factor from both sides get cancelled but conceptually it is wrong and can result in loss of marks. - Formulation of equations with the accurate RHS and LHS should be done in context of multiple systems before getting into substitution and result along with the significance of each term in the equation.
(ii)(a)	• Most candidates answered it correctly as 'X'. However, some answered as 'Y' due to the lack of clarity of the concept. • Some candidates misinterpreted the relationship and considered that a material exhibiting a greater rise in temperature has a higher specific heat capacity.	- Explain clearly that low specific heat capacity material will result in a greater rise in temperature and vice versa if mass remains constant. - Include problems where students need to compare the temperature changes of different materials, given that the heat input and masses are same. - Explicitly state the common misconception (higher temperature rise means higher SHC) by referring to the formula and the concept of specific heat capacity.
(ii)(b)	• Most of the candidates could answer this part correctly. • Some candidates who got it wrong misunderstood the inverse relationship between rise in temperature and specific heat capacity for a given heat input.	- Emphasise that a substance with low specific heat capacity shows a rapid and high rise in temperature and reinforce the concept by giving suitable examples.
(ii)(c)	• Many candidates answered it correctly.	- Explain the concept by using kinetic theory of molecules with the help of conversion of ice into water.
(iii)(a)	• Many candidates correctly answered this question by successfully applying 'Fleming's Left-Hand Rule'. • A significant number of candidates answered this question incorrectly showing lack of clarity of the rule.	- Give practice problem based on finding the force experienced by a current carrying conductor when placed in a magnetic field. - Acquaint students with the concept by changing the orientation of the diagram.
(iii)(b)	• Many could identify the name correctly, but some candidates gave the incorrect answers.	- Clearly explain the difference between the Fleming's right hand rule and Fleming's left hand rule and when to use them.

<table>
<tr><th colspan="2">Comments of Examiners</th><th>Suggestions for teachers</th></tr>
<tr><td></td><td>• Some of the incorrect names written were-
➢ Fleming’s right hand rule.
➢ Right hand rule.
➢ Thumb rule.
➢ Faraday’s law.
➢ Lenz’s law.</td><td>- Provide and discuss variety of questions based on Fleming’s left hand rule.
- Encourage students to make a list of laws used in electromagnetism and tabulate them with their application for revision.</td></tr>
<tr><td>(iii)(c)</td><td>• Most candidates were unable to understand the concept correctly.
• Some candidates applied the formula $F = BIl\sin\theta$, and answered ‘no effect’ as the expression does not include area of cross-section.
• Only a few candidates were able to answer it correctly. Most of them could not connect to the idea of more current with increased cross section and hence more force.</td><td rowspan="2">- Encourage step-by-step analysis in solving a problem.
- Explain that the effect of physical quantities that may not explicitly present in a mathematical expression can indirectly influence the outcome.
- Discuss the factors that influence the force experienced by a conductor when placed in a magnetic field.
- Make the students aware of the interconnection between various concepts and bring this connection while discussing magnetism through different examples.</td></tr>
<tr><td>(iii)(d)</td><td>• Many candidates failed to answer this part of the question correctly.
• As the concept was not clear, the candidates only verified the factors such as magnetic field, current and the length of the conductor that are given in the expression.</td></tr>
</table>

GENERAL COMMENTS

Topics found difficult by candidates	
	• Centre of gravity of an irregular lamina. • Application of law of conservation of energy in numerical. • Change of resistivity with temperature. • Resistance of a circuit when one bulb is removed. • Numerical on moment of force. • Application of Lenz's law. • Applications of total internal reflection. • Diagram of block and tackle. • Graphical representation of emission of alpha and beta particles. • Current rating of a pole fuse. • Direction of magnetic field lines. • Drawing ray-diagrams based on total internal reflection. • Resonance of sound.

Concepts in which candidates got confused	
	• Colour coding of wires. • Analysing condition for total internal reflection. • Application of Lenz's law. • Energy conversion. • Graphical representations. • Resistance and Resistivity. • Series and parallel combination. • Position of element in the periodic table after alpha emission. • Position of centre of gravity. • Direction of magnetic field lines inside and outside the magnet. • Identifying poles of a magnet. • Relation between e.m.f. and terminal volage of an electrical cell.

Suggestions for candidates

- Practice examples, derivations, and proofs, and try to explain concepts in your own words.
- Solve a wide variety of problems, from basic to challenging, to strengthen your understanding of fundamental principles and develop problem-solving skills.
- Focus on understanding the 'why' behind formulas and concepts, not just memorising them.
- Relate concepts to real-world situations for better understanding.
- Practice numerical problems regularly from the textbook and other resources provided by the CISCE to reinforce concepts and become familiar with the types of questions asked.
- Solve competency-focused practice questions, previous years' board papers, and the specimen paper issued by CISCE.
- Pay attention to writing correct units for all physical quantities.
- Learn principles, laws, rules, and definitions accurately.
- Practice ray diagrams regularly to improve precision and clarity.
- Draw neat diagrams while solving numerical problems on moments of force and heat to ensure step-by-step understanding.
- Recheck all calculations carefully after solving each problem.
- Review previous years' Pupil Performance Analysis to understand common mistakes made by candidates.
- Take practice tests under timed conditions to simulate the exam environment.
- Read the question paper carefully during the first fifteen minutes of reading time.
- Keep your handwriting neat and legible.
- Always draw arrows on rays when attempting questions that require ray diagrams.
- Attempt the questions in the same order as given in the question paper.
- Manage your time effectively while writing the paper.
- Recheck your calculations and steps before submitting your answer sheet.

Suggestions for Teachers

- Discussion of theories, principles and mathematical formulation in physics should be planned around strategies that enhance the Capacity for Scientific inquiry in the students.
- It is important to be creative when it comes to designing diverse scenarios connected to real life (as far as possible) to discuss various concepts in physics.
- Alternate trajectories to approach and solve problems should be addressed and encouraged among the students.
- Mathematical formulation and substitution are the usual factors of fear while considering physics as a subject; rather than dictating steps, the students should be given time to discuss in groups and find the rationale for the equations and mathematical steps involved in various concepts.
- Connections across concepts should be emphasized wherever possible to give better clarity and utility of certain ideas. Encourage students to build on their previous ideas and not approach concepts in isolation.